

Deep learning model to detect deceptive generative adversarial network generated images using multimedia forensic

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ABSTRACT

Computer-generated imagery has been made more lifelike and misleading by new Generative Adversarial Network (GAN) models, such as StyleGAN, posing severe risks to people's safety, social order, and privacy. Deceptive content creation, including Deepfakes, image tampering, and information hiding, can be facilitated through the misuse of GANs. To tackle these challenges, a detection model is proposed in this research, employing a spatial-frequency joint dual-stream convolutional neural network. Learnable frequency-domain filtering kernels and frequency-domain networks are leveraged to thoroughly learn and extract frequency-domain features, considering the discernible artifacts left by GAN images in the frequency spectrum due to the upsampling process during production. Lastly, the two sets of traits are combined to identify GAN-created faces. The proposed model outperforms state-of-the-art methods, as evidenced by experimental findings on various datasets, both in terms of detection accuracy on high-quality created datasets and generalization across datasets.

1. Introduction

In recent years, the rapid progress in internet technology and artificial intelligence have ushered in a transformative era of information transmission, making digital images the primary medium due to their rich content and ease of storage. These images have widespread applications across diverse fields, including finance, healthcare, and news media. However, this technological surge has also led to potential threats, as malicious actors exploit these innovations to create deceptive content, such as Deepfakes, image tampering, and information hiding [1,2].

The accessibility of advanced technologies enables users to effortlessly manipulate facial expressions, synthesize virtual voices, and even swap faces with celebrities, offering novel and enjoyable experiences. However, alongside the benefits come significant risks.

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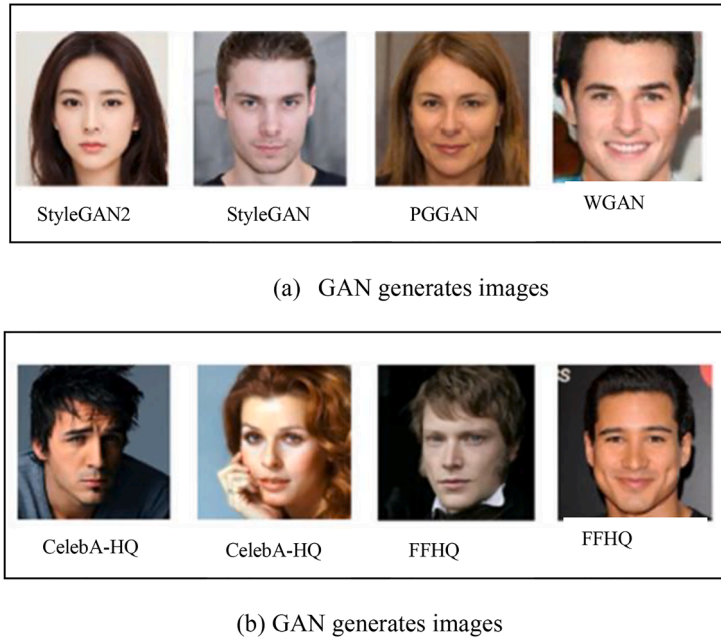


Fig. 1. GAN-generated images and real images.

Malicious individuals use these technologies to alter images with harmful intent, disseminating them widely online. This violates personal privacy, disrupts social stability, and seriously harms national security and interests [3].

The multimedia field has been captivated by the success of Generative Adversarial Networks (GANs), particularly with the introduction of the Progressive Growing of GANs (PGGAN) by Karras and others at the renowned international conference ICLR in 2018. PGGAN employs a layer-by-layer training approach, enabling the generation of high-resolution images up to 1024×1024 , effectively addressing the issue of low-resolution image generation in GANs. These generated images exhibit remarkable resolution and appear smooth and realistic, devoid of apparent artefacts. In Fig. 1, we showcase various counterfeit facial images created by GANs alongside samples from multiple actual facial image datasets.

Nonetheless, as the saying goes, "Seeing is no longer synonymous with truth." The rise of sophisticated image manipulation techniques has given rise to unprecedented threats to digital image authenticity and integrity. Researchers have tirelessly analyzed GAN structures to identify anomalous traces left during image generation, leading to the development of effective methods for detecting forged facial images. The field of Multimedia forensics has actively explored solutions to combat the challenges posed by deepfakes and deceptive image manipulation [4]. Researchers leverage cutting-edge technologies and interdisciplinary expertise to safeguard the trustworthiness of digital images and mitigate the potentially harmful consequences of misinformation and image-based attacks.

However, research has found that these algorithms exhibit good detection performance in simple scenarios [5]. Yet, when facing images generated by unknown GAN models or undergoing various image editing operations, such as JPEG compression, down-sampling, or Gaussian noise, the detection performance of these algorithms significantly declines [6]. Furthermore, existing algorithms often treat spatial domain features and frequency domain features as two separate domains and independently use spatial domain or frequency domain images as detection criteria, failing to combine them to explore the distribution characteristics of GAN-generated images fully.

The objective of this work is to overcome the difficulties associated with producing realistic and high-resolution false photos using GANs (Generative Adversarial Networks), particularly in the context of facial images. In order to ensure the authenticity and integrity of digital photos in diverse contexts, the study seeks to construct a reliable and efficient detection model that can properly discern between GAN-generated facial images and real photographs. In comparison to existing techniques, the proposed model achieves higher detection performance by utilizing a dual-stream convolutional neural network that incorporates both spatial and frequency domain features.

Deepfake technology, which makes it possible to create realistic-looking but fake multimedia content, has raised questions about potential abuse. Researchers are working hard to create techniques for identifying and lessening the effects of Deepfakes in order to solve this problem. H. Chen et al.'s [7] innovative framework for deepfake detection focuses on learning features of intra-consistency and inter-diversity, capturing key differences between real and fake content, and resulting in generalizable deepfake detection that can handle various Deepfake variations and evasive tactics.

This research suggests a detection approach based on a dual-stream convolutional neural network that combines the spatial and frequency domains to overcome the aforementioned issues. The main goal is to jointly investigate the key distinctions between GAN-generated images and actual photos in order to swiftly and precisely determine the veracity of test samples. The model develops

learnable frequency domain filtering kernels and a frequency domain network to completely learn and extract frequency domain features since GAN images may have checkerboard artifacts in the frequency domain as a result of up sampling operations during the generation process. Additionally, to lessen the effects of information loss during the transformation to the frequency domain, a spatial domain network is created to learn the distinct spatial domain properties of image content. In the end, both sets of attributes are used to detect facial images produced by GAN. The suggested model performs better than existing methods in terms of accuracy, generalization, and robustness, according to experimental findings. The main contribution of work:

- 1 In order to effectively distinguish between GAN-generated facial images and actual photographs, the paper suggests a unique detection approach based on a dual-stream convolutional neural network that incorporates spatial and frequency domain characteristics.
- 2 In order to thoroughly analyze the frequency domain properties of GAN-generated images and improve detection performance, the model combines learnable frequency domain filtering kernels and a frequency domain network.
- 3 A spatial domain network is included to learn distinct spatial domain properties of visual content, which helps to increase the accuracy of detection and reduce information loss during transformation to the frequency domain.
- 4 Even when presented with images produced by unidentified GAN models or subjected to various image editing techniques, the new model outperforms previous methods in terms of accuracy, generalization, and robustness.
- 5 The research helps to improve the security and integrity of digital photographs by offering a reliable way to identify fake facial images produced by GANs and defending against possible dangers to personal privacy and national security.

1.1. Organization of the paper

The present section introduces the concept of GAN-Images and previously implemented techniques and their limitations. [Section 2](#) describes the Related Work in the field of GAN-Generated Facial Image Detection. [Section 3](#) describes the proposed Spatial-Frequency Integrated Gan Deepfake Image Detection Model. [Section 4](#) describes the experimental results found, discussion on the results and comparative results with previous techniques. The last section summarizes the findings as a conclusion regarding the present work and future scope of the research.

2. Related work

2.1. GAN-Generated facial image detection algorithms

Existing representative algorithms for GAN image detection can be broadly categorized into two types: those based on spatial domain information [3,5,6] and those relying on frequency domain information [4,7-9]. Spatial domain-based algorithms can further be divided into those utilizing handcrafted features and those leveraging convolutional neural networks (CNNs) for detection.

On the other hand, Chen et al. [7] was among the first to use CNN for detecting GAN-generated facial images. While achieving the detection of forged images through supervised learning, this method is highly susceptible to adversarial attacks. To address this, Mimura [8] extracts co-occurrence matrices from the RGB channels of images and stacks them along the channel dimension, inputting them into a CNN for classification. However, this algorithm overlooks the correlations between colour channels. Building upon this research, Özkanoglu and Ozer [9] proposes a new model called Cross-Net, utilizing three forms of cross-colour channel combinations: RB, RG, and GB, and computing cross-colour channel co-occurrence matrices based on these combinations. Experimental results show that Cross-Net enhances the robustness of the detection algorithm compared to using individual colour channel information alone.

Furthermore, to improve robustness against image editing operations and the threat of unknown GAN generation models, researchers have introduced advanced ideas from computer vision. For instance, Lei et al. [10] proposes a dual-stream CNN detection algorithm, where one stream uses Gaussian low-pass filters to compute low-frequency components, and the other uses high-pass filters to calculate high-frequency residuals. These two streams are fused and input into the final fully connected layer for classification, enhancing the model's robustness.

Addressing frequency domain-based methods, the Hu et al. [11] observes that GAN-generated images exhibit anomalies in their high-frequency components due to repeated upsampling during the generation process. Based on this insight, they propose a detection method based on DCT (Discrete Cosine Transform) frequency domain transformation. Similarly, Chen et al. [7] designs a detection algorithm that inputs image colour channel spectra and employs capsule networks as the core architecture, achieving good results. Additionally, Guo et al. [12] analyzes the regression relationship between image pixels, considering the generation process of GAN images as a signal processing procedure using a set of finite impulse response filters. By characterizing the distribution of the most significant DCT coefficients, they achieve good detection accuracy for GAN image anomalies.

While these algorithms exhibit good detection performance in simple scenarios, their performance significantly declines when faced with images generated by unknown GAN models or undergoing various image editing operations. Therefore, this research focuses on improving the generalization and robustness of the algorithms and designing a universal detection model applicable to complex scenarios and diverse data.

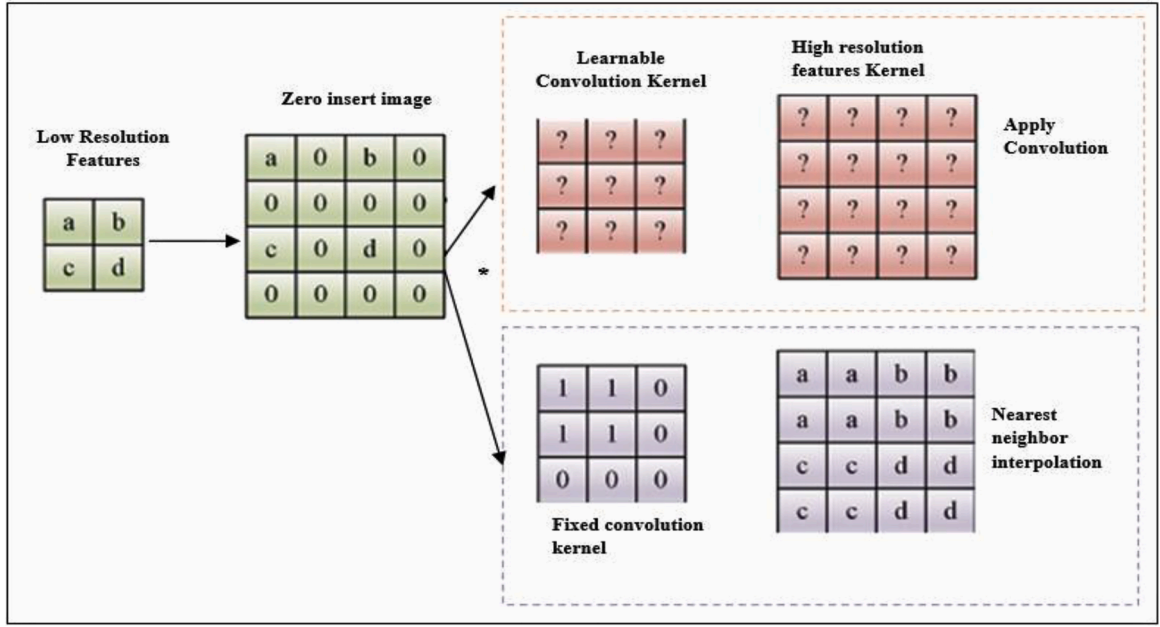


Fig. 2. Up sampling operations (Transpose Convolution and Nearest-Neighbor Interpolation).



Fig. 3. Amplitude spectra of low-resolution image and zero-inserted image.

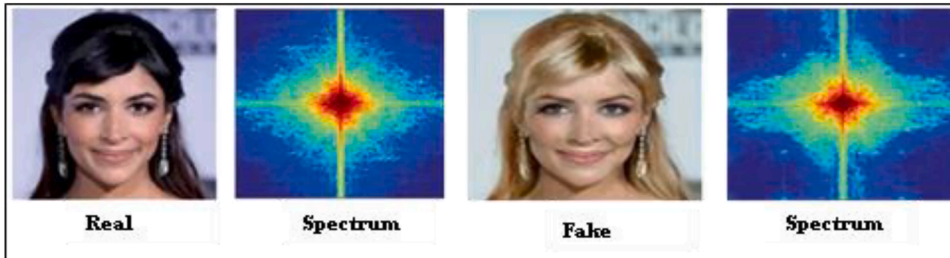


Fig. 4. Amplitude spectra of real and generated images (In the electronic version, the figure is in color).

2.2. Up sampling artifacts in GAN images

During the image generation process in GAN, the generator performs upsampling operations to gradually increase the image resolution, an essential step in image generation. Different GAN models employ different up sampling operations [13]. For example, DC-GAN [14] uses transpose convolution (also known as de-convolution) for upsampling, while PGGAN [15] utilizes nearest-neighbor interpolation. Fig. 2 illustrates the details of these two operations, where $*$ represents convolution.

The up sampler enhances the horizontal and vertical resolutions by a factor of m from a low-resolution feature tensor input. This

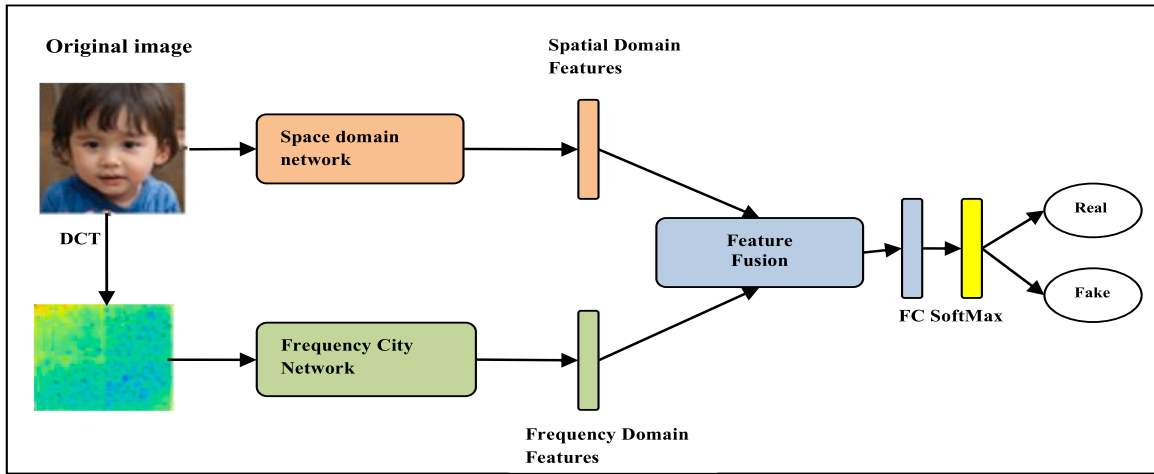


Fig. 5. spatial-frequency integrated GAN Deepfake image detection model.

paper assumes $m = 2$, which is the most typical setting, for demonstrative purposes. The low-resolution feature tensor is up sampled by inserting a zero row/column after each row/column, and convolution is then used to provide appropriate values to the newly inserted places. The learnable nature of the convolution kernel in transpose convolution, as opposed to the fixed values used in nearest-neighbor interpolation, is what distinguishes it from nearest-neighbor interpolation.

In transpose convolution, the up-sampling artifacts are known as "checkerboard artifacts." Hu et al. [11] gave a thorough analysis in the frequency domain, while Adiban et al. [16] explored this in the spatial domain. The amplitude spectrum of the final high-resolution image is formed by multiple copies of the spectrum of the original low-resolution image being duplicated in the high-frequency portion of the image, as shown in Fig. 3 in accordance with the characteristics of the Discrete Fourier Transform (DFT). The low-frequency portion of the spectrum is shifted to the centre for illustration's sake.

It is obvious that the high-frequency components of the picture spectrum have anomalies after interpolation. The high-frequency components must be eliminated or diminished in order to prevent these artifacts from appearing in the final generated image. As a result, the convolution kernels shown in Fig. 3 after that are often low-pass filters. Although it cannot be ensured that the convolution kernels in transpose convolution are strictly low-pass, checkerboard artifacts can still be seen in many photos because the convolution kernels in transpose convolution are learnable. As seen in Fig. 4, the two image on the left show a real face image and its associated amplitude spectrum, whereas the images on the right show a face created by a GAN and its associated spectrum. The checkerboard effects in the frequency domain are represented by the red-highlighted areas in the generated image's spatial domain, and in the amplitude spectrum's corresponding bright spots enclosed by green ellipses. These artifacts were produced by the generator's up-sampling module. Since fixed low-pass filter convolution kernels better reduce the artifacts, the nearest-neighbor up sampler is used. The forging traces are not entirely eliminated, though, and if the low-pass filter removes too much high-frequency information from the final image, the visual quality may suffer, making it simpler to identify from genuine images.

Based on the above theoretical analysis, this study aims to analyze real and GAN-generated images in the frequency domain and use the image amplitude spectrum as input for the network to learn features. Additionally, this study combines the frequency domain features learned with spatial domain features to propose a GAN face detection model based on spatial-frequency integration, which will be detailed in Section 3.

3. Spatial-frequency integrated GAN deepfake image detection model

Based on the analysis of the frequency domain artifacts in natural and GAN-generated images mentioned earlier, this study constructs a frequency domain network to extract and learn the image's spectral features. As the conversion from spatial to frequency domain is a double-edged sword, it can concentrate the energy of the face image, but at the same time, it discards many components representing fine details. To maximize the effectiveness of the model, this study also builds a spatial domain network to learn the differences in image content. Finally, the features learned separately are merged to form the spatial-frequency integrated GAN Deepfake image detection model, as illustrated in Fig. 5.

3.1. Construction of the frequency domain network

As analyzed in Section 2.2, during the image generation process, the GAN generator performs upsampling operations that introduce unique artifacts in the frequency domain. These artifacts differ from those present in natural images, as evident from Figs. 3 and 4. The upsampling not only introduces anomalies in the high-frequency part of the spectrum but also causes a checkerboard effect, leading to uneven distribution throughout the entire spectrum.

To improve the recognition efficiency of real and fake images, we aim to build a specialized network capable of fully learning and

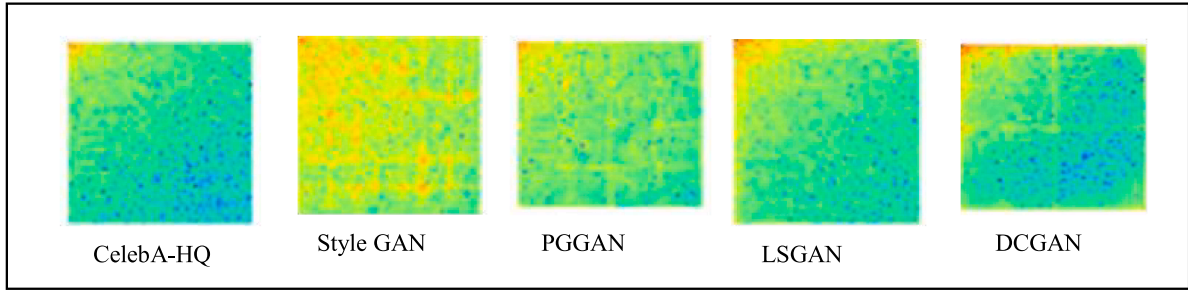


Fig. 6. Average amplitude spectra of real images and images generated by different GAN Models.

addressing these frequency domain artifacts. Such artifacts pose challenges for traditional image recognition methods, necessitating a tailored approach. Our proposed network will focus on learning these distinctive characteristics to enhance the overall performance and accuracy of real vs. fake image detection.

Incorporating the suggested comments, the section now provides a concise explanation of the upsampling artifacts' impact and the specific purpose of the proposed network. This strengthens the clarity and significance of the research approach.

3.1.1. Image transformation from spatial domain to frequency domain

To perform feature learning on face images in the frequency domain, this study utilizes Discrete Cosine Transform (DCT) to transform the images from the spatial domain to the frequency domain. Originally used for processing one-dimensional data (such as text and speech), DCT has found wide application in image processing due to its energy concentration properties. There are currently 8 forms of Discrete Cosine Transform, and this study uses the most common form, DCT-II, with the following calculation:

$$G(v, w) = 4 \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} T(x, y) \cos \frac{(2x+1)v\pi}{2M} \cos \frac{(2y+1)w\pi}{2N} \quad (1)$$

$$1 \leq u \ll M, 1 \leq v \ll N, T(x, y) \in [0, 255]$$

here, T and G represent the image in the spatial and frequency domains, respectively, and the image size is $M \times N$. $T(x, y)$ represents the pixel value at coordinate (x, y) in the image's spatial domain, and $G(v, w)$ represents the frequency component at coordinate (v, w) in the image's frequency domain.

Fig. 6 shows the average amplitude spectra of real images and images generated by different GAN models. Although each GAN model has a different structure, they all leave similar artifacts in the frequency domain, namely, the clearly visible checkerboard artifacts in each image. Next, this study builds a network to fully learn the frequency domain artifacts, aiming to improve the efficiency of recognizing real and fake images.

3.1.2. Learnable filter kernel

After the image is transformed from the spatial domain to the frequency domain, certain characteristics (such as noise and spatial information redundancy) become more prominent and easier to manipulate. Therefore, this study takes advantage of these characteristics in the frequency domain to further analyze face images. Since image filtering is achieved by element-wise multiplication between matrices, calculations in the frequency domain are very efficient. This study plans to utilize element-wise multiplication as a frequency domain filtering method to achieve effective feature extraction. For example, common high-pass and low-pass filters can be used to retain high-frequency information, highlighting image details and generating a sharpened image, or to retain low-frequency information and remove image noise, resulting in a smoother image. However, to filter out unnecessary information and extract the desired image features, targeted filter design is necessary. To efficiently generate appropriate filters, this study uses learnable frequency domain filter kernels (LFK), which act as trainable filters for feature extraction. With the property of backpropagation in convolutional neural networks, the filter kernels can be repeatedly learned and automatically adjust their multiple weights within the network. Compared to manually setting parameter values, this approach is more concise, efficient, and can extract the most effective facial information. The definition of a single filter kernel is as follows:

$$L = \begin{bmatrix} Z_{11} & \cdots & Z_{1n} \\ \vdots & \ddots & \vdots \\ Z_{m1} & \cdots & Z_{mn} \end{bmatrix} \quad (2)$$

In the filter kernel L , $Z(x, y)$ represents the trainable weight, whose values are randomly initialized based on a truncated normal distribution with a mean of 0 and a standard deviation of 0.1. The size of the filter kernel is $m \times n$, which is the same as the input image. Since the filter kernel can automatically adjust its weights through training in the network, it is very efficient and time-saving. In contrast, existing methods mostly rely on prior knowledge and hand-designed approaches to set the weights of the filter kernel, which may not extract the most effective facial information and can waste a considerable amount of time.

The frequency domain feature map referred to in this study has the same form as the feature map in a convolutional neural network, both represented in matrix form. The difference lies in one containing image frequency domain information, while the other represents

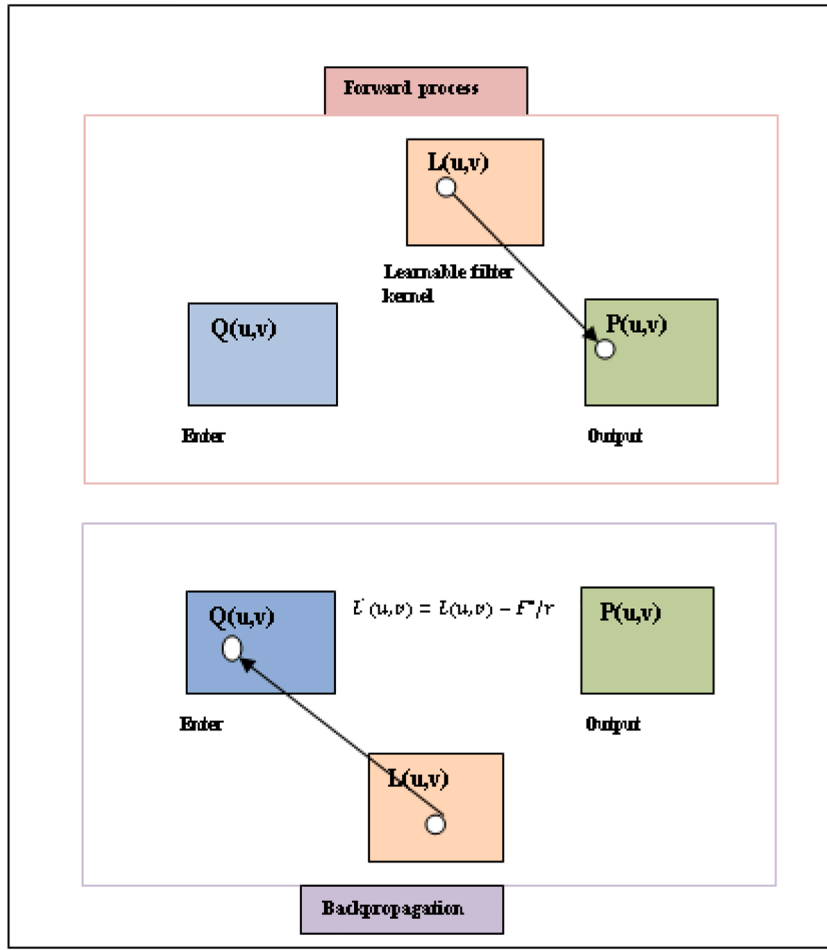


Fig. 7. Learnable filter kernel.

spatial domain information of the image. When the image is transformed into the frequency domain and input into the network, the filter kernel performs frequency domain filtering on the input through element-wise multiplication, and the resulting output feature map has the same size as the original image. The specific process of the learnable filter kernel is shown in Fig. 7.

In Fig. 7, Q represent the input spectrum of the image, L is the filter kernel with the same size as the input, and P is the output feature map. The forward process of this filtering is defined as follows:

$$P(u, v) = Q(u, v) \cdot L(u, v) \quad (3)$$

where $1 \leq u \leq m, 1 \leq v \leq n$, and m and n are the number of rows and columns of the input matrix Q , respectively. For a single element point in the output feature map, it is calculated using a single multiplication operation. Compared to the weighted sum of multiple points in the convolutional filtering, this calculation is more efficient, which is one of the advantages of traditional frequency domain image processing.

The definition of the backward propagation process for this filtering is as follows:

$$L'(u, v) = L(u, v) - G^* / lr \quad (4)$$

Among them, F is the gradient, and lr is the learning rate. Iteratively adjusts its own weight to extract the most effective face information.

3.1.3. Frequency domain network construction

The frequency domain network aims to extract discriminative features from natural and real images in the frequency domain. This study introduces a four-part structure, consisting of the pre-processing stage, feature extraction layer, feature aggregation layer, and final feature classification.

In the feature extraction layer, learnable filter kernels are utilized as tools for image feature extraction. Each input feature spectrum

Table 1

Frequency domain network structure and parameter configuration.

Network structure	Variable	Data dimension	Parameter settings
Preprocessing	-	256×256	None
feature extraction	Filt ₁	256×256×32	K1=32
	Filt ₂	256×256×32	K2=1
	Filt ₃	256×256×32	K3=1
Feature Aggregation	Conv1	256×256×32	Conv(3×3×32)
	Pool1	128×128×32	Maxpool(2×2)
	Conv1	128×128×48	Conv(3×3×48)
	Pool1	64×64×48	Maxpool(2×2)

Table 2

Spatial Domain Network Structure and Parameter Configuration.

Data variable	Data dimension	Parameter settings
Enter	256×256	None
conv₁₁	256×256×24	Conv(3×3×24)
conv₁₂	256×256×24	Conv(3×3×24)
pool₁	128×128×24	Maxpool(2×2)
conv₂₁	128×128×36	Conv(5×5×36)
conv₂₂	128×128×36	Conv(5×5×36)
pool₂	64×64×36	Maxpool(2×2)
conv₃	64×64×48	Conv(7×7×48)
pool₃	32×32×48	Maxpool(2×2)

is processed with K frequency domain filter kernels. The value of L, known as the feature map expansion coefficient, is specifically set as listed in Table 1. The role of L is to increase the number of feature maps for a more comprehensive extraction of information embedded in the spectrum. Notably, n represents the total number of feature extraction layers, where the filter denotes the ith feature extraction layer. Li is carefully chosen as the expansion coefficient for each layer to generate sufficient feature maps. It is essential to strike a balance in selecting n; a value too small might not extract enough features, while an excessively large value could lead to a time-consuming model with convergence issues. Hence, based on subsequent experiments, this study sets n as 3.

The feature aggregation layer, mainly composed of convolution and pooling operations, enhances the interrelation between output feature maps and achieves dimension reduction. Convolution is initially employed to locally aggregate frequency information,

Table 3

Forged face and real face datasets used in the experiment.

Authenticity	Name	Resolution	Quantity
Fake Face Dataset	StyleGANII	1024×1024	8000
	StyleGANI	1024×1024	8000
	PGGAN	256×256	10000
	WGAN	256×256	10000
	BEGAN	128×128	10000
	LSGAN	128×128	10000
Real Face Dataset	CelebA-HQ	1024×1024	12000
	CelebA-HQ	256×256	12000
	CelebA-HQ	128×128	12000

followed by pooling operations to reduce feature dimensions while preserving as much image information as possible, thereby accelerating model convergence. This study adopts max pooling for dimension reduction since low-frequency information in the image's frequency domain contains the vast majority of relevant data and exhibits large absolute values.

The detailed configuration of the frequency domain network is provided in Table 1, though it would benefit from additional insights into the reasoning behind specific hyperparameter choices. Moreover, a comparative analysis with existing image feature extraction and classification methods would strengthen the study's contributions. Addressing potential limitations and discussing how they can be mitigated in future work would further enhance the paper's robustness and practical applicability.

3.2. Spatial domain network construction

In Section 3.1, the frequency domain network was developed to extract image spectrum artifacts for classification. Although this leverages energy concentration in the discrete cosine transform and reduces computational complexity, it discards image details, affecting detection performance. Moreover, different GAN models have varying upsampling frequencies and artifact sizes, making it challenging to accurately discriminate poorly generated images. To address this, we explore the spatial domain, constructing a spatial domain network for feature extraction.

The joint spatial-frequency model achieves improved GAN face image detection results, with a three-layer design. Inspired by VGG, our model utilizes a dual-layer convolutional structure to expand the receptive field of convolution kernels, complemented by the non-saturating LReLU activation function. Refer to Table 2 for the spatial domain model's structure and parameter configuration.

3.3. Joint spatial-frequency dual-stream CNN network

In Sections 3.1 and 3.2, the frequency domain and space domain networks are respectively constructed to extract their respective feature information. To better realize the task of true and false face detection, this paper fuses the two features to form a final dual-stream CNN model based on a space-frequency combination.

After convolution and pooling operations, the frequency and space domain feature lengths are different and cannot be directly fused. Therefore, this paper first performs dimension conversion to ensure that the weights of these two features are equal during feature fusion. Specifically, through full connection, the two features are converted into parts with a length of 1024, and the two parts are fused by point-by-point addition. Finally, this paper sends the fused elements to a fully connected layer composed of 512 neuron units and uses the unsaturated function LReLU for activation. L2 regularization is also carried out in the fully connected layer, where the parameter λ is 0.0005. Since this paper studies the binary classification problem for detecting true and false face images, the Soft-max layer generates classification probabilities and performs the final classification.

Table 4
Results of ablation experiments.

Precision	StyleGAN	PGGAN	WGAN
FreNet	99.68	98.87	95.2
SpaNet	98.59	98.68	99.68
SpaGFreNet	99.86	99.49	99.77

4. Experimental setup and results analysis

In this section, the experimental effectiveness of the proposed algorithm is evaluated, incorporating details about the datasets and evaluation metrics. A comprehensive set of experiments is conducted to demonstrate the efficacy of the proposed joint spatial-frequency detection model. The model's performance is assessed on both globally generated GAN Deepfake and locally generated GAN face images, showcasing robust detection performance. Two diverse datasets are used for evaluation purposes. The first dataset comprises globally generated GAN Deepfake face images sourced from various publicly available repositories, ensuring a broad representation of Deepfake techniques and variations. The second dataset encompasses locally generated GAN face images obtained from controlled experiments, enabling assessment of the model's performance in more challenging and realistic scenarios.

4.1. Datasets and evaluation metrics

For the fake face data set, this paper selects images generated by six different structures of GAN networks, including the advanced GAN models Style-GANII [13] and StyleGANI [14] with high generation quality, the mid-term generation models PGGAN [2] and WGAN [15], and the early models BEGAN [16] and LSGAN [17–18] whose generation quality is not clear enough. For authentic face images, this paper uses the CelebA-HQ data set [19–22]. This paper selects three resolutions to match the face images generated by GAN. The resolutions and numbers of all data sets are listed in Table 3.

The reason for such diverse datasets is to ensure that the trained model can adapt to more practical and broader needs. The model needs to be effective not only for images generated by multiple structured GANs but also for images generated by different resolution GANs of different structures. Since the input image size for this dual-stream model is 256×256 , all images are resized to 256×256 . In all experiments, the dual-stream CNN is trained only on StyleGAN I and CelebA HQ datasets. The real face images are maintained at a resolution of 1024×1024 to match StyleGAN I, while the other five GAN-generated face datasets are used solely for testing the generalization and robustness of the trained model. Finally, the performance of the model is measured using accuracy (ACC) with the following formula:

$$\text{Accuracy} = \frac{U_p + U_N}{Q + N} \quad (5)$$

where Q and N represent the number of generated samples and real samples in the dataset, and U_p and U_N represent the number of correctly detected generated samples and correctly detected real samples, respectively.

4.2. Experimental results and analysis

4.2.1. Ablation experiment

The resolution of the test datasets includes 1024×1024 , 256×256 , and 128×128 . After training, the model is evaluated on a test set containing 3000 real face images and 3000 forged face images. The experimental results are shown in Table 4. The second row of the table represents the accuracy of the individual frequency domain network (FreNet) in classifying real and forged faces. The third row shows the accuracy obtained by the individual spatial domain network (SpaNet). The fourth row displays the face authentication accuracy achieved by fusing the frequency domain features with the spatial domain features (SpaFreNet). It is evident from the table that the proposed approach based on the joint spatial-frequency network model achieved the highest classification accuracy in all test datasets, confirming its effectiveness in GAN face detection.

Furthermore, as shown in Table 4, the standalone FreNet achieved remarkable accuracy for StyleGAN I, which produces high-quality and extremely realistic faces. However, for early GAN models such as WGAN, the accuracy is only 95.20 %. This is due to StyleGAN models going through multiple up sampling operations during the image generation process, resulting in higher image resolution. The multiple up sampling also leaves clear and distinguishable artifacts in the high-frequency components of the images, which can be effectively detected by the frequency domain network alone. On the other hand, for datasets like WGAN, which have lower image clarity, using only the frequency domain network for classification is not ideal, leading to many misclassifications. This is a crucial reason for designing the spatial domain network. By complementing the discarded detail information in the frequency domain

Table 5
Analysis of the number of feature extraction layers.

Extraction Layer Number of Layers	Precision
1	65
2	97.55
3	99.86
4	99.88
5	-

Table 6
Generalization experiment comparison of different algorithms on StyleGANI,BeGAN, and LsGAN generated datasets.

Generated Datasets			
Test set	StyleGANII	PGGAN	WGAN
Literature [19]	92.97	98.04	99.53
Literature [20]	94.37	97.64	99.26
Literature [21]	98.05	97.45	95.25
Literature [22]	98.56	97.37	97.06
Proposed Model	99.24	98.67	98.9

Table 7
Generalization experiment comparison of different algorithms on StyleGAN II, PGGAN, and WGAN generated datasets.

Generated Datasets			
Test set	StyleGANII	PGGAN	WGAN
Literature [19]	94.41	98.45	99.89
Literature [20]	93.28	98.77	99.75
Literature [21]	98.78	98.27	95.63
Literature [22]	98.91	98.22	96.02
Proposed Model	99.36	98.9	98.83

Table 8
Results of robustness testing.

Model	Compression	Tailoring	Vague	Noise	Combination
Literature [19]	75.76	90.78	78.67	53.26	70.22
Literature [20]	95.24	94.08	93.45	95.26	95.01
Literature [21]	94.83	98.83	93.61	89.56	92.27
Literature [22]	95.04	98.24	95.77	91.23	94.96

and performing synchronous extraction and learning in the spatial domain, the spatial-frequency joint model achieves ideal detection performance.

4.2.2. Frequency domain feature extraction layer analysis

The frequency domain network's depth and training convergence are influenced by the number of feature extraction layers (n). In our comparative experiments, we tested different values of n (1, 2, 3, 4, 5) and recorded the results in Table 5. With $n = 1$, the network's feature learning was limited, leading to a classification accuracy of approximately 65 %. However, as n increased to 2, the model demonstrated gradual improvement, significantly boosting the accuracy. For $n = 3$ or 4, the model achieved exceptional performance with close to 100 % accuracy on the dataset. Yet, when $n = 5$, convergence became problematic due to excessive depth. After careful consideration, we selected $n = 3$ as it provided desired accuracy and faster convergence compared to $n = 4$, optimizing the training time.

4.2.3. Generalization experiment

As the research progresses, the development of GAN networks will continue to produce more and higher quality new models. To truly apply the algorithm in practice, the ability to generalize is a crucial evaluation criterion. Therefore, we tested the generalization ability of the spatial-frequency joint GAN face detection model to see if it can achieve satisfactory results when facing unknown GAN-generated images.

We conducted generalization experiments using images generated by six different GAN structures, as shown in Tables 6 and 7. In Table 6, three sets of forged images were used for training, derived from StyleGAN I, BeGAN, and LsGAN. For each GAN, we randomly selected 4000 images for training, resulting in a total of 12,000 forged face images. Correspondingly, we selected 12,000 real face images from the CelebA HQ dataset, including three different resolutions, making the total number of training images 24,000. For the remaining three GANs (StyleGAN II, PGGAN, and WGAN), we used the pre-trained model mentioned above for testing and selected real face images that match their respective resolutions.

We conducted generalization experiments by exchanging the positions of the first set of GAN-generated datasets used for training and testing, and the results are shown in Table 7. This means using images generated by StyleGAN II, PGGAN, and WGAN to train the model and testing the model's performance using images generated by StyleGAN I, BeGAN, and LsGAN.

To better illustrate the advantages of the proposed approach, we compared the model's performance with some GAN detection algorithms [16]. Overall, the proposed model shows remarkable generalization performance, which confirms that even though each GAN model has different structures, as long as upsampling operations are used, frequency domain artifacts will be left behind. Once the model learns to detect these artifacts, its performance in identifying other GAN-generated images can be improved. Most notably, the proposed spatial-frequency joint structure significantly enhances the detection rate of StyleGAN, which produces the highest quality forged images. The classification results reach over 99 %. Compared to high-resolution images, datasets with lower clarity like LsGAN are less likely to reveal high-frequency features. However, with the assistance of spatial domain features, the classification accuracy is still maintained at a good level.

4.2.4. Robustness experiment

To assess whether the artifacts used for image discrimination, such as those left in the frequency domain, remain effective after post-processing operations when images are uploaded to the internet or downloaded to mobile phones, we evaluated the robustness of the proposed spatial-frequency joint detection model against common image disturbances. These disturbances include compression, cropping, blurring, random noise, and combinations of these disturbances. The specific settings are as follows:

- 1) Images were subjected to JPEG compression, and the selected compression quality factor was randomly sampled from the range [10, 90].
- 2) Images were randomly cropped, with cropping percentages randomly sampled from the range [5, 20]. The cropped images were then upsampled to their original resolution.
- 3) Images were subjected to Gaussian blurring, with kernel sizes randomly sampled from the range [0.5, 2.5].
- 4) Images were affected by Gaussian noise, with variance randomly sampled from the range [5, 20].
- 5) All the above perturbations were combined.

In this experiment, both the training and testing datasets were derived from CelebA HQ (real) and StyleGAN I (forged) datasets, and the testing model used was the one saved in the first column and fourth row of Table 4. The results of the robustness experiment are shown in Table 8.

It is evident that the proposed spatial-frequency joint network achieved high accuracy of 99.68 % in handling image cropping, but its accuracy significantly declined under other image disturbances. The fifth row in Table 8 presents the detection performance of the standalone frequency domain network. Since the frequency domain network's core is to capture the chessboard artifacts in high-resolution image spectra for detection, it benefits more from using more upsampling operations and higher-resolution images for learning and feature extraction however, compression, blurring, and noise severely affected image clarity, indirectly undermining the clarity of chessboard artifacts in the frequency spectrum, leading to suboptimal model performance.

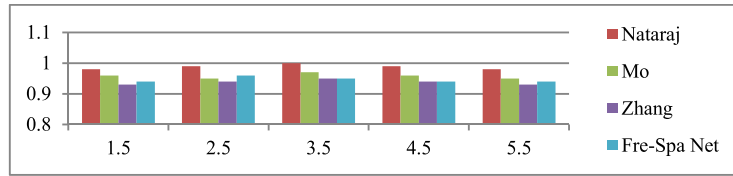
Notably, the addition of spatial domain features on top of frequency domain features significantly improved accuracy for each category, indicating that the approach of combining spatial and frequency domain features for image authentication is correct. However, future research should focus on further enhancing the model's robustness.



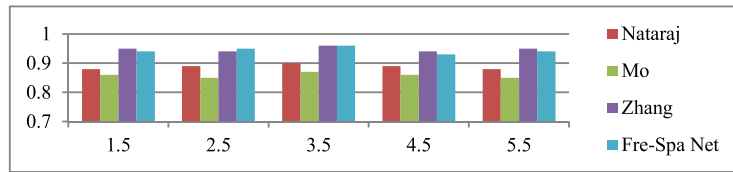
(a) Rule mask



(b) Irregular mask

Fig. 8. Some samples from the LGGF dataset.

(a)



(b)

Fig. 9. The detection accuracy for different forged region sizes.

4.2.5. Test on locally forged GAN faces

Currently, for GAN image detection, most attention is given to full-face images, with little focus on locally forged GAN faces. However, in reality, there are many cases where GANs are used for face restoration. Some are done with good intentions, but others may intentionally alter facial features such as eyes and expressions, distorting facts and causing negative consequences. Therefore, this experiment aims to explore whether the proposed model has a broader application scope. The real dataset used is from FFHQ, and the locally forged GAN face dataset is from LGGF [23–25]. The LGGF dataset contains 95,200 images with forged regions of varying shapes and sizes, as shown in Fig. 8.

The evaluation of different forged region sizes will be conducted, and the entire test set includes six sizes with proportions of generated images ranging from 0.5 % to 5.0 %. Fig. 9(a) presents the detection accuracy for regular forged regions, and Fig. 9(b) presents the detection accuracy for irregular forged regions. It is evident that the detection accuracy of all models increases with the increase in the forged region size. The proposed spatial-frequency joint model achieves the best results for both types of forged regions. This indicates that in locally forged GAN faces, the texture differences in spatial domain features and the high-frequency artifacts in frequency domain information still play a role in authenticating the images. This finding suggests broader applicability and future optimization considerations for the model.

5. Conclusion

This study proposed a novel detection strategy based on a spatial-frequency joint convolutional neural network, leveraging learnable filter kernels and a frequency domain network to identify chessboard artifacts in the frequency domain. Spatial-frequency features were then utilized to verify facial images by employing a spatial domain network to learn distinct image content features. Experimental results on high-quality produced datasets demonstrated that the suggested algorithm outperformed previous methods regarding detection accuracy and generalization, highlighting its potential for a wider range of applications. However, the model does have some limitations. The impact of noise, compression, and image blurring on the frequency domain artifacts and the model's

resilience was observed, posing challenges to its robustness. To further enhance the model's performance and address its drawbacks, future optimization efforts should focus on minimizing the effects of noise, compression, and image blurring. Additionally, exploring techniques to improve the model's adaptability to diverse image manipulations and variations will be essential for real-world deployment and comprehensive defense against deceptive content. Overall, this research contributes valuable insights into GAN image detection and offers a promising foundation for future advancements in multimedia forensics, aiding in the ongoing battle against the proliferation of deepfakes and deceptive image manipulation.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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